

Numbers, Groups and Codes

Section 1.1: Division Algorithm and Greatest Common Divisors

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Why study integers?

Integers are fundamental in:

- cryptography and secure communication
- QR codes and error detection
- computer algorithms
- reliable data transmission

This week: core tools used throughout the course.

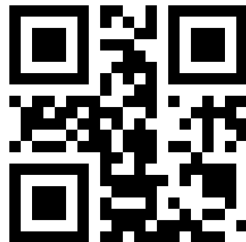


QR codes: a quick motivation

QR codes remain readable even if partially damaged.

Idea: add redundancy $\Rightarrow$ detect/correct errors.

Later in the course, this becomes *coding theory*.



Plan for today

- Division algorithm: existence + uniqueness + examples
- Divisibility notation
- Greatest common divisors: definition + properties
- Euclidean algorithm: computation + why it works
- Bézout identity: statement + back substitution
- Congruences: meaning + basic computations (preview)

Warm-up: division with remainder

Example:

$$175 = 11 \cdot 15 + 10.$$

- 15 is the **quotient**
- 10 is the **remainder**

Warm-up: quick practice

Find q, r such that $b = aq + r$ and $0 \leq r < a$:

$$93 = 7q + r, \quad 250 = 13q + r.$$

Work for 45 seconds, then compare with a neighbor.

Warm-up: answers

$$93 = 7 \cdot 13 + 2 \quad (q = 13, r = 2)$$

$$250 = 13 \cdot 19 + 3 \quad (q = 19, r = 3)$$

The Division Algorithm (Theorem)

Let $a, b \in \mathbb{Z}$ with $a > 0$. Then there exist integers q, r such that

$$b = aq + r, \quad 0 \leq r < a.$$

Moreover, q and r are **unique**.

Why require $0 \leq r < a$?

Without the restriction, many decompositions exist:

$$175 = 11 \cdot 15 + 10 = 11 \cdot 14 + 21 = 11 \cdot 16 - 1 = \dots$$

We choose the **standard remainder** so computations are consistent.

Uniqueness: the key step

Assume

$$b = aq + r = aq' + r' \quad \text{with} \quad 0 \leq r, r' < a.$$

Subtract:

$$a(q - q') = r' - r.$$

Uniqueness: finish

Since $0 \leq r, r' < a$, we have $-a + 1 \leq r' - r \leq a - 1$.

$a(q - q')$ is a multiple of a . The only multiple of a in the interval $[-a + 1, a - 1]$ is 0.

Therefore $q = q'$ and $r = r'$.

Negative numbers: example

We still insist on:

$$b = aq + r, \quad 0 \leq r < a.$$

Example: divide -23 by 7 . We need $-23 = 7q + r$ with $0 \leq r < 7$.

Try $q = -3$: then $7q = -21$ and $r = -2$ (not allowed).

Try $q = -4$: then $7q = -28$ and $r = 5$ (allowed).

$$-23 = 7(-4) + 5.$$

Practice (with negatives)

Find q, r with $0 \leq r < a$:

$$-45 = 8q + r, \quad -1 = 6q + r.$$

(Work for 1 minute.)

Practice: answers

$$-45 = 8(-6) + 3 \quad (q = -6, r = 3)$$

$$-1 = 6(-1) + 5 \quad (q = -1, r = 5)$$

Mini-summary: Division Algorithm

- Every integer b can be written as $b = aq + r$ with $0 \leq r < a$.
- q and r are unique.
- The rule matters most when b is negative.

Divisibility notation

$$3 \mid 12$$

Read: **“3 divides 12.”**

It also means:

- 12 is a multiple of 3;
- 12 is divisible by 3;
- 3 is a divisor of 12.

Divisibility definition

We say a divides b if there exists an integer k such that

$$b = ak.$$

We write $a \mid b$.

If a does not divide b , we write $a \nmid b$.

Divisibility examples

$$3 \mid 12 \quad \text{because } 12 = 3 \cdot 4$$

$$5 \nmid 12 \quad \text{since no integer } k \text{ satisfies } 12 = 5k$$

$$-3 \mid 12 \quad \text{since } 12 = (-3) \cdot (-4)$$

Quick check

Which statements are true?

- $6 \mid 42$
- $7 \mid 42$
- $8 \mid 42$

(Decide quickly; no calculations beyond basic division.)

Quick check: answers

- $6 \mid 42$ true ($42 = 6 \cdot 7$)
- $7 \mid 42$ true ($42 = 7 \cdot 6$)
- $8 \mid 42$ false

Greatest common divisor (definition)

For integers a, b not both zero, $\gcd(a, b)$ is the largest integer $d \geq 1$ such that

$$d \mid a \quad \text{and} \quad d \mid b.$$

How to read gcd

$$\gcd(12, 18) = 6$$

Read: **“the greatest common divisor of 12 and 18 is 6.”**

(We also say: “gcd of 12 and 18.”)

Examples

$$\gcd(12, 18) = 6 \quad \gcd(8, 15) = 1$$

If $\gcd(a, b) = 1$, we say a and b are **coprime**.

Coprime: another example

Are 14 and 25 coprime?

Yes, because their only common divisor is 1:

$$\gcd(14, 25) = 1.$$

Common divisors and linear combinations

If $d \mid a$ and $d \mid b$, then for any integers x, y ,

$$d \mid (ax + by).$$

This simple fact will reappear in Bézout identity.

Basic properties of gcd

For integers a, b and integer k :

- $\gcd(a, b) = \gcd(b, a)$
- $\gcd(a, b) = \gcd(a, b - a)$
- $\gcd(a, b) = \gcd(a, b + ka)$

Why $\gcd(a, b) = \gcd(a, b - ka)$?

If $d \mid a$ and $d \mid b$, then $d \mid (b - ka)$.

Conversely, if $d \mid a$ and $d \mid (b - ka)$, then $d \mid b$.

So the set of common divisors is unchanged.

Checkpoint

Which is easier to compute?

$$\gcd(84, 30) \quad \text{or} \quad \gcd(30, 84 - 2 \cdot 30)?$$

(Explain why they are the same.)

Goal: compute gcd efficiently

Listing divisors works for small numbers but fails for large ones.

We need an algorithm that uses division with remainder.

Euclidean Algorithm (procedure)

Assume $a > b > 0$. Compute remainders:

$$a = bq_1 + r_1, \quad 0 \leq r_1 < b$$

$$b = r_1q_2 + r_2, \quad 0 \leq r_2 < r_1$$

Continue until a remainder is 0.

Result: the last nonzero remainder is $\gcd(a, b)$.

Why it works (key identity)

If

$$a = bq + r,$$

then, by the basic properties of gcd,

$$\gcd(a, b) = \gcd(b, a) = \gcd(b, a - bq) = \gcd(b, r).$$

Worked example: $\gcd(252, 198)$

$$252 = 1 \cdot 198 + 54$$

$$198 = 3 \cdot 54 + 36$$

$$54 = 1 \cdot 36 + 18$$

$$36 = 2 \cdot 18 + 0$$

So the last nonzero remainder is 18:

$$\gcd(252, 198) = 18.$$

Practice: compute a gcd

Compute:

$$\gcd(420, 96).$$

Work in pairs (2–3 minutes). Write down each division step.

Answer: $\gcd(420, 96)$

$$420 = 4 \cdot 96 + 36$$

$$96 = 2 \cdot 36 + 24$$

$$36 = 1 \cdot 24 + 12$$

$$24 = 2 \cdot 12 + 0$$

So $\gcd(420, 96) = 12$.

Euclid's algorithm is fast even for huge numbers.

This is one reason it is used in cryptography and algorithms.

Bézout identity (statement)

For integers a, b not both zero, there exist integers x, y such that

$$\gcd(a, b) = ax + by.$$

Interpretation: the gcd is a **linear combination** of a and b .

How to find x, y : back substitution

Euclid's algorithm produces remainders as combinations of earlier numbers.

By substituting backwards, we can express the gcd as $ax + by$.

Example to back-substitute: $\gcd(171, 30)$

Euclid steps:

$$171 = 5 \cdot 30 + 21$$

$$30 = 1 \cdot 21 + 9$$

$$21 = 2 \cdot 9 + 3$$

So $\gcd(171, 30) = 3$.

Back substitution

$$21 = 2 \cdot 9 + 3 \quad \Rightarrow \quad 3 = 21 - 2 \cdot 9.$$

We need to express 9 and 21 using 171 and 30.

$$30 = 1 \cdot 21 + 9 \quad \Rightarrow \quad 9 = 30 - 21.$$

Substitute into $3 = 21 - 2 \cdot 9$:

$$3 = 21 - 2(30 - 21) = 3 \cdot 21 - 2 \cdot 30.$$

$$171 = 5 \cdot 30 + 21 \quad \Rightarrow \quad 21 = 171 - 5 \cdot 30.$$

Substitute:

$$3 = 3(171 - 5 \cdot 30) - 2 \cdot 30 = 3 \cdot 171 - 17 \cdot 30.$$

So $x = 3$, $y = -17$ works.

Practice: Bézout coefficients

Find integers x, y such that

$$\gcd(35, 12) = 35x + 12y.$$

(Work in pairs for 3–4 minutes.)

Answer: $\gcd(35, 12)$ and Bézout

Euclid:

$$35 = 2 \cdot 12 + 11, \quad 12 = 1 \cdot 11 + 1$$

Back-substitute:

$$1 = 12 - 11 = 12 - (35 - 2 \cdot 12) = 3 \cdot 12 - 35.$$

So

$$1 = (-1) \cdot 35 + 3 \cdot 12.$$

Mini-summary: gcd/EUCLID/Bézout

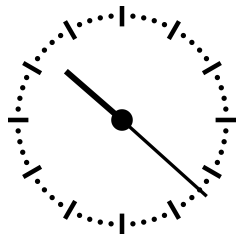
- Euclid computes $\gcd(a, b)$ efficiently.
- Back substitution expresses $\gcd(a, b)$ as $ax + by$.
- This is a key tool for solving modular equations (later).

Clock arithmetic (intuition)

On a 12-hour clock:

$$17 \equiv 5 \pmod{12}.$$

Numbers differing by multiples of 12 represent the same time.



How to read congruence

$$17 \equiv 5 \pmod{12}$$

Read: **“17 is congruent to 5 modulo 12.”**

(Informal: “same remainder when divided by 12.”)

Definition of congruence

We write

$$a \equiv b \pmod{n}$$

if n divides $a - b$:

$$n \mid (a - b).$$

Equivalently: a and b leave the same remainder when divided by n .

Examples: compute remainders

Compute:

$$23 \bmod 7, \quad 58 \bmod 9.$$

Then decide whether:

$$35 \equiv 5 \pmod{10}.$$

$$23 = 7 \cdot 3 + 2 \Rightarrow 23 \bmod 7 = 2$$

$$58 = 9 \cdot 6 + 4 \Rightarrow 58 \bmod 9 = 4$$

And $35 \equiv 5 \pmod{10}$ because $10 \mid (35 - 5) = 30$.

Congruence classes (pattern)

Numbers congruent to 2 (mod 5):

$$\dots, -8, -3, 2, 7, 12, 17, 22, \dots$$

They differ by multiples of 5.

Given integers a and n with $n \geq 1$, the **congruence class of a modulo n** is the set

$$[a]_n := \{b \in \mathbb{Z} : b \equiv a \pmod{n}\}.$$

We have

$$\dots = [-8]_5 = [-3]_5 = [2]_5 = [7]_5 = [12]_5 = \dots$$

Looking ahead

Next week we will solve equations like

$$7x \equiv 1 \pmod{26}.$$

This is a key step behind encryption.

Summary of Section 1.1

- Division algorithm: existence and uniqueness of $b = aq + r$
- Divisibility: reading and meaning of $a \mid b$
- gcd: definition and basic properties
- Euclid's algorithm computes gcd efficiently
- Bézout identity: $\gcd(a, b) = ax + by$
- congruence: same remainder / divisibility of differences